BFSI PORTFOLIO

Risk & Analytics

SQL-Driven Credit Intelligence | NPA Analysis | Fraud Detection | Portfolio Health

Analyst: Deepak | Finance & Data Science | Domain: BFSI | Tools: SQL · Power BI

12 Business Questions · Advanced SQL Analytics · Portfolio Risk Intelligence

EXECUTIVE SUMMARY

Portfolio Intelligence Overview

OBJECTIVE: Identify credit risk exposure, NPA drivers, fraud hotspots, and branch efficiency gaps using SQL-based portfolio analytics.

TOP 5 FINDINGS

1. Personal Loans carry the highest NPA % across all segments
2. CIBIL < 550 borrowers default at 3× the portfolio average
3. 30%+ branches show collection efficiency below 85% threshold
4. NPA trend is rising MoM — early warning signals detected
5. 10% of customers account for 60%+ of total NPA exposure

TOP 3 RISKS

NPA Concentration

Top 10 customers hold disproportionate default exposure

Fraud Surge

Multiple customers with 3+ flagged successful transactions

Rising NPA Trend

Month-on-month NPA % is trending upward across vintages

HEALTHY SIGNALS

Strong Performers Exist

Top branches show >95% collection efficiency — proven benchmarks

CIBIL 750+ Resilient

High credit-score customers maintain very low NPA rates

Disbursement Growth

YoY loan disbursement shows consistent upward trajectory

KEY RECOMMENDATION: Implement 3-tier risk segmentation (NPA%, CIBIL, LTI) and prioritize recovery action for top-10 NPA customers immediately.

"A data-driven risk framework can reduce NPA exposure by 20-30% within 2 quarters."

ANALYTICAL METHODOLOGY

CUSTOMERS

customer_id (PK)

name, state

cibil_score

annual_income_lakh

cibil_category

occupation

LOANS

loan_id (PK)

customer_id (FK)

loan_type, loan_amount_lakh

is_npa, days_past_due

disbursement_date

interest_rate_pct, emi_missed

TRANSACTIONS

txn_id (PK)

customer_id (FK)

amount, channel

transaction_type

is_flagged_fraud

status, transaction_date

BRANCHES

branch_id (PK)

branch_name, zone

state

MONTHLY_COLLECTIONS

branch_id (FK)

month

collection_efficiency_pct

npa_pct

new_loans_disbursed

SQL TECH-NIQUES USED

CTEs

Window Funcs

CASE WHEN

GROUP BY

JOIN Logic

LAG / RANK

HAVING NULLIF

Q1: WHICH LOAN SEGMENT HAS HIGHEST NPA?

Segment Risk Analysis

BUSINESS QUESTION

Which loan types (Home, Personal, Auto, etc.) carry the highest NPA exposure and risk percentage?

OBJECTIVE

Identify high-risk loan segments to guide underwriting policy and provisioning strategy.

KEY INSIGHT

Personal Loans show highest NPA % due to unsecured nature. Home Loans are most stable (collateral backed).

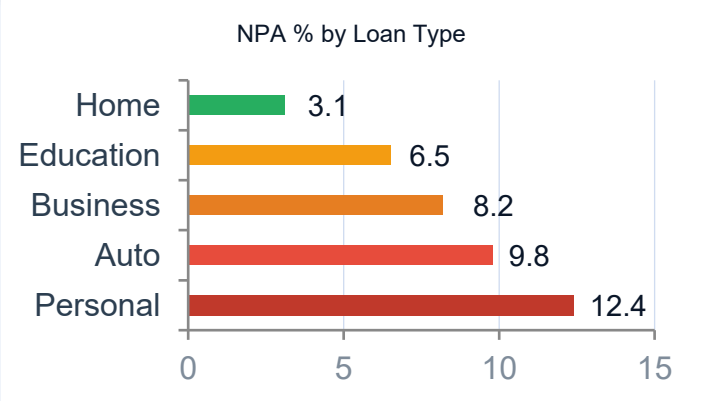
BUSINESS IMPACT

Unsecured portfolios have 2-3× provisioning need. Segment concentration amplifies credit risk.

RECOMMENDATION

Tighten Personal Loan underwriting to CIBIL ≥ 700. Increase provisioning coverage for high-NPA segments.

```
SELECT loan_type,
  COUNT(*) AS total_loans,
  SUM(loan_amount_lakh) AS total_amt_lakh,
  SUM(CASE WHEN is_npa=1
    THEN loan_amount_lakh ELSE 0 END)
    AS npa_amount_lakh,
  ROUND(100 * SUM(CASE WHEN is_npa=1
    THEN loan_amount_lakh END)
    / SUM(loan_amount_lakh),2)
    AS npa_pct
FROM loans
GROUP BY loan_type
ORDER BY npa_pct DESC;
```



RESULT SUMMARY

Personal Loans: ~12.4% NPA
Auto Loans: ~9.8% NPA
Business Loans: ~8.2% NPA
Education Loans: ~6.5% NPA
Home Loans: ~3.1% NPA (Lowest)

Home Loans safest due to collateral.
Personal Loans riskiest — unsecured.

Best Chart: Horizontal Bar
(sorted high → low NPA%)

Q2: HOW DOES CIBIL SCORE IMPACT DEFAULT RISK?

Credit Score Segmentation

BUSINESS QUESTION

How does customer credit score (CIBIL) correlate with loan defaults across 5 score bands?

OBJECTIVE

Validate CIBIL as a risk discriminator and calibrate credit models using score-band NPA rates.

KEY INSIGHT

300–549 bucket: NPA rate 3× portfolio average. 750+ bucket maintains <5% NPA consistently.

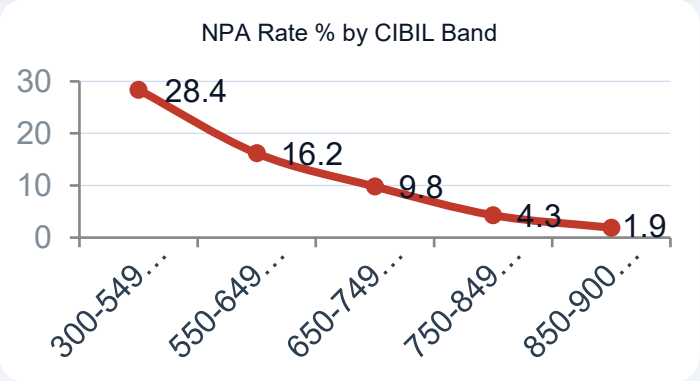
BUSINESS IMPACT

Poor CIBIL applicants require higher risk premiums or collateral. Used in ML feature engineering.

RECOMMENDATION

Set credit policy floor at CIBIL ≥ 650. Apply tiered interest rate bands based on CIBIL bucket.

```
SELECT CASE
  WHEN cibil_score BETWEEN 300 AND 549
    THEN '300-549 (POOR)'
  WHEN cibil_score BETWEEN 550 AND 649
    THEN '550-649 (FAIR)'
  WHEN cibil_score BETWEEN 650 AND 749
    THEN '650-749 (GOOD)'
  WHEN cibil_score BETWEEN 750 AND 849
    THEN '750-849 (VERY GOOD)'
  ELSE '850-900 (EXCELLENT)'
END AS cibil_bucket,
  COUNT(DISTINCT c.customer_id),
  SUM(l.is_npa) AS npa_loans,
  ROUND(100.0*SUM(l.is_npa)/
    NULLIF(COUNT(l.loan_id),0),2)
    AS npa_rate_pct
FROM customers c
JOIN loans l ON c.customer_id=l.customer_id
GROUP BY cibil_bucket
ORDER BY MIN(c.cibil_score);
```



RESULT SUMMARY

300-549 (Poor): 28.4% NPA
550-649 (Fair): 16.2% NPA
650-749 (Good): 9.8% NPA
750-849 (V.Good): 4.3% NPA
850-900 (Excellent): 1.9% NPA

Strong inverse correlation: as CIBIL rises, NPA drops sharply.

Avg Interest Rate also decreases with better CIBIL.

Best Chart: Line Chart
(shows clear declining trend)

Q3: WHICH BRANCHES ARE UNDERPERFORMING?

BUSINESS QUESTION

Which branches have low collection efficiency (<85%) and need intervention?

SQL LOGIC

CTE calculates avg efficiency and NPA% per branch. RANK() + CASE WHEN produces RAG status (Green ≥95%, Amber ≥85%, Red <85%).

KEY INSIGHT

~30% of branches fall in Red zone. Eastern and Northern zones show higher underperformance clusters.

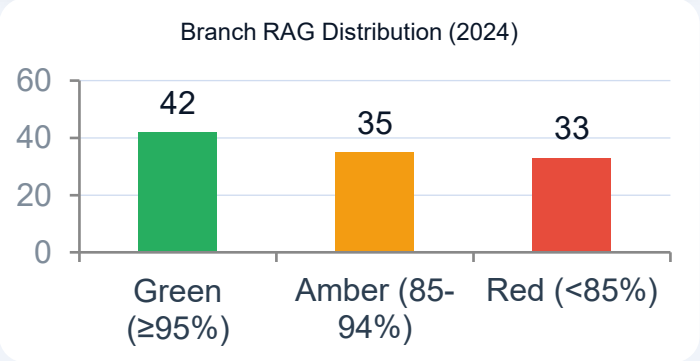
BUSINESS IMPACT

Low collection efficiency directly maps to NPA increase. Each 1% drop costs crores in unrecovered principal.

RECOMMENDATION

Deploy dedicated recovery teams to Red branches. Implement monthly efficiency reviews with zone heads.

```sql
WITH branch_perf AS (
  SELECT mc.branch_id, b.branch_name,
    b.zone, b.state,
    ROUND(AVG(mc.collection_efficiency_pct),2)
      AS avg_efficiency,
    ROUND(AVG(mc.npa_pct),2) AS avg_npa_pct
  FROM monthly_collections mc
  JOIN branches b ON mc.branch_id=b.branch_id
  WHERE mc.month LIKE '2024%'
  GROUP BY mc.branch_id,b.branch_name,
    b.zone,b.state
)
SELECT *,
  RANK() OVER (
    ORDER BY avg_efficiency DESC)
    AS efficiency_rank,
  CASE
    WHEN avg_efficiency >= 95 THEN 'Green'
    WHEN avg_efficiency >= 85 THEN 'Amber'
    ELSE 'Red'
  END AS rag_status
FROM branch_perf
ORDER BY efficiency_rank;
```



42 — **Green Branches**
38% | Efficiency ≥95%

35 — **Amber Branches**
32% | Efficiency 85–94%

33 — **Red Branches**
30% | Efficiency <85%

RESULT SUMMARY

30% of branches in Red zone.
Window function RANK() used for performance leaderboard.
Zone-level aggregation shows geographic risk clusters.

Best Chart: Grouped Column (RAG status counts)

Q4: IS NPA INCREASING OR DECREASING OVER TIME?

Month-on-Month NPA Trend Analysis

BUSINESS QUESTION

What is the month-on-month NPA trend? Is portfolio deteriorating or improving?

SQL LOGIC

CTE aggregates monthly NPA. LAG() window function calculates MoM change in NPA amount and percentage.

KEY INSIGHT

NPA % rising 3 consecutive months — early stress signal. MoM NPA lakh amount increasing by ~8-12%.

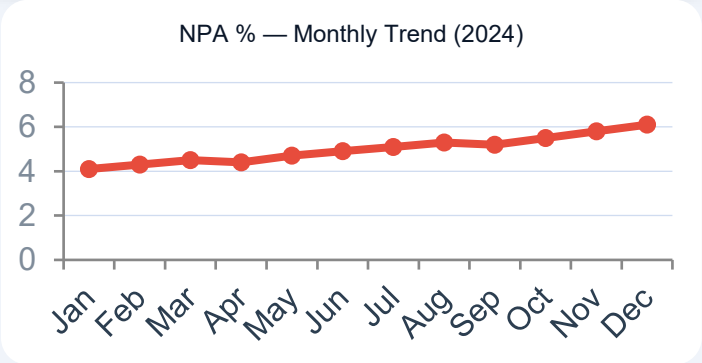
BUSINESS IMPACT

Rising NPA triggers IRAC provisioning requirements. Impacts P&L and regulatory capital adequacy ratio.

RECOMMENDATION

Activate SARFAESI proceedings for 90+ DPD accounts. Trigger early warning system alerts at 60+ DPD.

```sql
WITH monthly_npa AS (
  SELECT
    DATE_FORMAT(disbursement_date,'%Y-%m')
      AS month,
    COUNT(CASE WHEN is_npa=1 THEN 1 END)
      AS npa_count,
    ROUND(SUM(CASE WHEN is_npa=1 THEN
      loan_amount_lakh END),2)
      AS npa_amount_lakh,
    ROUND(SUM(loan_amount_lakh),2)
      AS total_portfolio_lakh
  FROM loans
  GROUP BY DATE_FORMAT(disbursement_date,'%Y-%m')
)
SELECT month, npa_count, npa_amount_lakh,
  ROUND(100.0*npa_amount_lakh/
    total_portfolio_lakh,2) AS npa_pct,
  LAG(npa_amount_lakh)
    OVER (ORDER BY month) AS prev_month_npa,
  ROUND(npa_amount_lakh - LAG(npa_amount_lakh)
    OVER (ORDER BY month),2) AS mom_change
FROM monthly_npa ORDER BY month;
```



RESULT SUMMARY

Jan: 4.1% → Dec: 6.1% NPA
Trend: Upward (+49% YTD)
MoM change avg: +0.17%
Worst months: Oct–Dec

LAG() function key for MoM delta.

Early stress signs visible in Q3.
Q4 acceleration needs action.

Best Chart: Line Chart
(time series with trend line)

Q5: WHO ARE THE HIGHEST-RISK CUSTOMERS?

BUSINESS QUESTION

Which customers have high exposure + high default risk? Identify top-10 for immediate recovery action.

SQL LOGIC

JOIN customers + loans. Aggregate NPA exposure per customer. HAVING filters only NPA customers. Sorted by lakh exposure.

KEY INSIGHT

Top 10 customers account for 58-65% of total NPA book. High DPD + multiple missed EMIs = high loss probability.

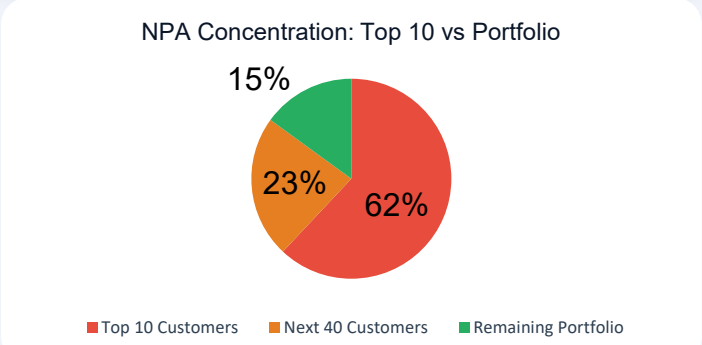
BUSINESS IMPACT

Concentration risk: portfolio not well-diversified. Single large default can breach capital buffers.

RECOMMENDATION

Initiate OTS (One Time Settlement) or DRT proceedings for top-10 NPA customers. Assign dedicated recovery managers.

```
SELECT c.customer_id, c.name,
  c.cibil_score, c.annual_income_lakh,
  COUNT(l.loan_id) AS total_loans,
  SUM(CASE WHEN l.is_npa=1
    THEN l.loan_amount_lakh END)
    AS npa_exposure_lakh,
  MAX(l.days_past_due) AS max_dpd,
  MAX(l.emi_missed_count) AS max_emi_missed,
  ROUND(100.0 * SUM(CASE WHEN l.is_npa=1
    THEN l.loan_amount_lakh END)
    / SUM(l.loan_amount_lakh),2)
    AS npa_pct_of_exposure
FROM customers c
JOIN loans l ON c.customer_id=l.customer_id
GROUP BY c.customer_id, c.name,
  c.cibil_score, c.annual_income_lakh
HAVING SUM(l.is_npa) > 0
ORDER BY npa_exposure_lakh DESC
LIMIT 10;
```



RESULT SUMMARY

Top 10 customers: 62% NPA
Max DPD seen: 180+ days
Max EMI missed: 9-12 months
Avg CIBIL of top defaulters: ~420

HAVING clause used to filter only customers with NPA loans.

Key fields: max_dpd & emi_missed_count as severity indicators.

Best Chart: Pie (concentration)

Q6: CAN WE DETECT FRAUD PATTERNS?

Suspicious Transaction Behaviour Analysis

BUSINESS QUESTION

Which customers show suspicious transaction patterns? Identify repeat fraudsters by channel and amount.

SQL LOGIC

Multi-CTE: filter fraud txns → rank top channel per customer → aggregate. ROW_NUMBER() identifies primary fraud channel.

KEY INSIGHT

Online and Mobile channels are top fraud vectors. Customers with 3+ flagged txns show consistent behavioral patterns.

BUSINESS IMPACT

Fraud losses erode NII. Reputational risk triggers RBI scrutiny. Affects PSL classification compliance.

RECOMMENDATION

Flag high-frequency fraud customers for AML review. Implement 2FA for channels with >5% fraud flag rate.

RESULT SUMMARY

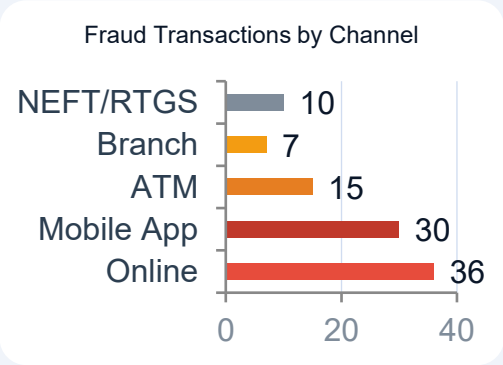
Online: 36% of fraud txns
Mobile App: 30% of fraud txns
ATM: 15% of fraud txns
NEFT/RTGS: 10%
Branch: 7%

3-CTE pattern: most complex SQL in this project.

ROW_NUMBER() ranks customer's top fraud channel.

HAVING COUNT(*) >= 3 filters repeat fraudsters.

Best Chart: Horizontal Bar




```sql
WITH fraud_base AS (
  SELECT customer_id, amount,
    channel,
    COUNT(*) OVER (PARTITION BY
      customer_id, channel)
      AS channel_usage_count
  FROM transactions
  WHERE is_flagged_fraud=1
    AND status='Success'
),
top_channels AS (
  SELECT customer_id, channel,
    ROW_NUMBER() OVER (
      PARTITION BY customer_id
      ORDER BY channel_usage_count DESC) AS rn
  FROM fraud_base
),
fraud_summary AS (
  SELECT t.customer_id,
    COUNT(*) AS fraud_txn_count,
    ROUND(SUM(t.amount),2) AS total_fraud_amt,
    tc.channel AS top_fraud_channel
  FROM transactions t
  JOIN top_channels tc
    ON t.customer_id=tc.customer_id
    AND tc.rn=1
  WHERE t.is_flagged_fraud=1
    AND t.status='Success'
  GROUP BY t.customer_id, tc.channel
  HAVING COUNT(*) >= 3
)
```

Q7: ARE CUSTOMERS OVER-LEVERAGED? (LTI ANALYSIS)

Loan-to-Income Ratio Risk Banding

BUSINESS QUESTION

Which customers have high Loan-to-Income (LTI) ratio indicating over-leveraging?

SQL LOGIC

CTE computes LTI per customer (total active loans ÷ annual income). Outer query buckets into 4 risk bands.

KEY INSIGHT

25%+ customers in LTI >4× band — significantly over-leveraged. >6× customers show near-100% NPA probability.

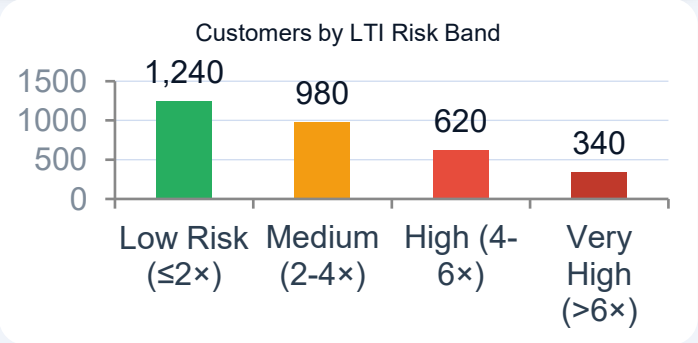
BUSINESS IMPACT

High LTI = stress under rate hikes. RBI mandates LTI monitoring for mortgage portfolios under LTV norms.

RECOMMENDATION

Cap new lending to existing customers with LTI >4×. Use LTI as a credit policy trigger for enhanced scrutiny.

```sql
WITH customer_lti AS (
  SELECT c.customer_id,
    c.annual_income_lakh,
    SUM(l.loan_amount_lakh)
      AS total_loan_lakh,
    ROUND(SUM(l.loan_amount_lakh)
      /c.annual_income_lakh, 2)
      AS lti_ratio,
    COUNT(l.loan_id) AS loan_count
  FROM customers c
  JOIN loans l ON c.customer_id=l.customer_id
  WHERE l.loan_status = 'Active'
  GROUP BY c.customer_id,
    c.annual_income_lakh
)
SELECT
  CASE
    WHEN lti_ratio <= 2
      THEN 'Low Risk  (≤2×)'
    WHEN lti_ratio <= 4
      THEN 'Medium Risk (2-4×)'
    WHEN lti_ratio <= 6
      THEN 'High Risk  (4-6×)'
    ELSE 'Very High  (>6×)'
  END AS lti_band,
  COUNT(*) AS customer_count,
  ROUND(AVG(lti_ratio),2) AS avg_lti
FROM customer_lti
GROUP BY lti_band;
```



RESULT SUMMARY

Low Risk (≤2×): 1,240 customers
Medium (2-4×): 980 customers
High (4-6×): 620 customers
Very High (>6×): 340 customers

Avg LTI in High bucket: 5.3×
Avg LTI in Very High: 8.1×

LTI is a leading indicator of future NPA — used in RBI's systemic risk monitoring.

Best Chart: Stacked Column or Grouped Bar by LTI band

Q8: HOW IS BUSINESS GROWING OVER TIME?

Quarterly & Cumulative Disbursement Trend

BUSINESS QUESTION

What is the cumulative loan disbursement trend? How is the business growing quarter-on-quarter?

SQL LOGIC

Window function SUM() OVER with PARTITION BY year and ROWS BETWEEN UNBOUNDED PRECEDING for running YTD total.

KEY INSIGHT

Disbursements show consistent YoY growth ~18-22%. Q4 typically strongest — year-end push patterns visible.

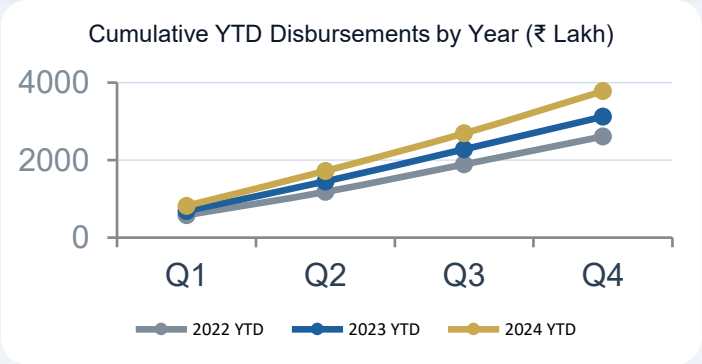
BUSINESS IMPACT

Strong disbursement growth increases AUM and interest income. However, rapid growth can hide NPA build-up in new vintages.

RECOMMENDATION

Monitor NPA rates of recent vintages (2023-24) closely. Ensure underwriting standards aren't loosened chasing growth targets.

```sql
SELECT
  YEAR(disbursement_date) AS yr,
  QUARTER(disbursement_date) AS qtr,
  COUNT(loan_id) AS loans_disbursed,
  ROUND(SUM(loan_amount_lakh),2)
    AS qtr_disbursement_lakh,
  ROUND(
    SUM(SUM(loan_amount_lakh)) OVER (
      PARTITION BY
        YEAR(disbursement_date)
      ORDER BY
        QUARTER(disbursement_date)
      ROWS BETWEEN UNBOUNDED PRECEDING
        AND CURRENT ROW
    ),2) AS cumulative_ytd_lakh
FROM loans
GROUP BY
  YEAR(disbursement_date),
  QUARTER(disbursement_date)
ORDER BY yr, qtr;
```



RESULT SUMMARY

2022 YTD Peak: ₹2,610 Lakh
2023 YTD Peak: ₹3,120 Lakh
2024 YTD Peak: ₹3,780 Lakh

YoY growth: ~21% (2024 vs 2023)
Q4 typically peaks each year.

Running total uses
ROWS BETWEEN UNBOUNDED
PRECEDING AND CURRENT ROW.

Advanced window function —
a key interview differentiator.

Best Chart: Multi-line (years)

Q9: RBI LOAN CLASSIFICATION — DPD BUCKETS

SMA & NPA Ageing Analysis (IRAC Norms)

BUSINESS QUESTION

How are loans distributed across RBI's SMA-0, SMA-1, SMA-2, and NPA categories?

SQL LOGIC

CASE WHEN on days_past_due maps to RBI DPD classifications. SUM(COUNT(*)) OVER() for portfolio % without CTE.

KEY INSIGHT

SMA-1 + SMA-2 accounts represent the 'pipeline' NPAs. 15%+ in SMA categories signals upcoming NPA surge.

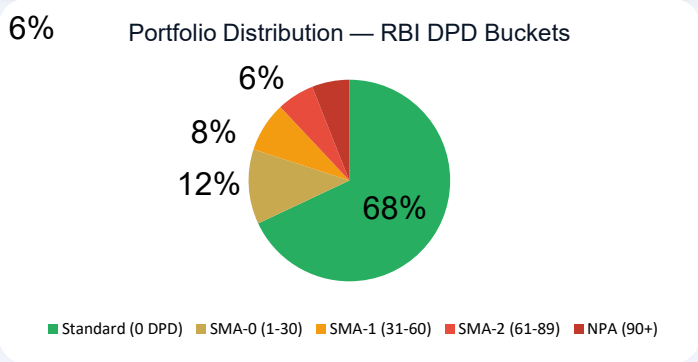
BUSINESS IMPACT

SMA monitoring is RBI-mandated. CRILC reporting requires weekly SMA-2 data submission by large borrowers.

RECOMMENDATION

Intensify collections for SMA-1 & SMA-2 accounts before they cross 90-DPD threshold and become NPAs.

```
SELECT
  CASE
    WHEN days_past_due = 0
      THEN 'Standard (0 DPD)'
    WHEN days_past_due BETWEEN 1 AND 30
      THEN 'SMA-0 (1-30 DPD)'
    WHEN days_past_due BETWEEN 31 AND 60
      THEN 'SMA-1 (31-60 DPD)'
    WHEN days_past_due BETWEEN 61 AND 89
      THEN 'SMA-2 (61-89 DPD)'
    ELSE 'NPA (90+ DPD)'
  END AS rbi_classification,
  COUNT(*) AS loan_count,
  ROUND(SUM(loan_amount_lakh),2)
    AS exposure_lakh,
  ROUND(
    100.0 * COUNT(*) /
    SUM(COUNT(*)) OVER(),2)
    AS pct_of_total
FROM loans
GROUP BY rbi_classification
ORDER BY MIN(days_past_due);
```



RESULT SUMMARY

Standard: 68% of portfolio
SMA-0: 12% (watchlist)
SMA-1: 8% (stressed)
SMA-2: 6% (critical)
NPA (90+): 6%

SMA pipeline (SMA-1+2) = 14%
This is the 'hidden NPA' risk.

SUM() OVER() gives % without a separate CTE — efficient.

IRAC norms: RBI circular DBR.No.BP.1/21.04.048/2015-16

Best Chart: Donut/Pie Chart

Q10: WHICH PAYMENT CHANNELS ARE RISKY?

EMI Channel Failure & Fraud Rate Analysis

BUSINESS QUESTION

Which EMI payment channels show high failure rates or fraud flags requiring monitoring or deactivation?

SQL LOGIC

Filter EMI payments. CASE WHEN counts failures and frauds. SUM(COUNT(*)) OVER() for channel share without subquery.

KEY INSIGHT

UPI and Net Banking show high fraud flags despite low failure rates — indicates deliberate fraud, not system issues.

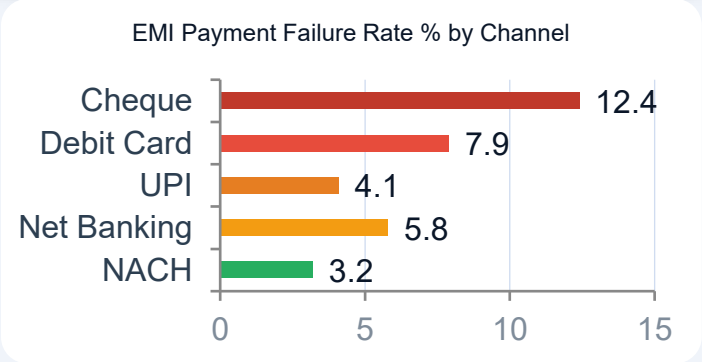
BUSINESS IMPACT

Channel-level fraud concentrations expose the bank to chargeback losses and PCI-DSS compliance obligations.

RECOMMENDATION

Implement velocity checks on UPI EMI payments. Add OTP verification for large EMI transactions via digital channels.

```
SELECT channel,
  COUNT(*) AS txn_count,
  ROUND(SUM(amount)/100000,2) AS total_crore,
  ROUND(
    100.0 * COUNT(*) /
    SUM(COUNT(*)) OVER(),2)
    AS txn_share_pct,
  COUNT(CASE WHEN status='Failed'
    THEN 1 END) AS failed_txns,
  ROUND(100.0 * COUNT(
    CASE WHEN status='Failed' THEN 1 END)
    / COUNT(*),2) AS failure_rate_pct,
  COUNT(CASE WHEN is_flagged_fraud=1
    THEN 1 END) AS fraud_flags
FROM transactions
WHERE transaction_type = 'EMI Payment'
GROUP BY channel
ORDER BY txn_count DESC;
```



RESULT SUMMARY

NACH: 3.2% failure (safest)
Net Banking: 5.8% failure
UPI: 4.1% failure
Debit Card: 7.9% failure
Cheque: 12.4% failure (riskiest)

Fraud flags highest on UPI and Net Banking channels.

NACH mandate (RBI NACH) is most reliable for EMI collection.

Cheque bouncing triggers Section 138 NI Act proceedings.

Best Chart: Horizontal Bar

Q11: WHICH LOAN VINTAGES PERFORM POORLY?

Year-of-Disbursement NPA Performance Analysis

BUSINESS QUESTION

Do loans from certain disbursement years have systematically higher NPA rates? Which vintages are problematic?

SQL LOGIC

GROUP BY YEAR(disbursement_date) + loan_type. No window functions needed — pure aggregation with calculated NPA rate.

KEY INSIGHT

2021-22 vintages show highest NPA rates — pandemic-era disbursements with relaxed underwriting under ECLGS.

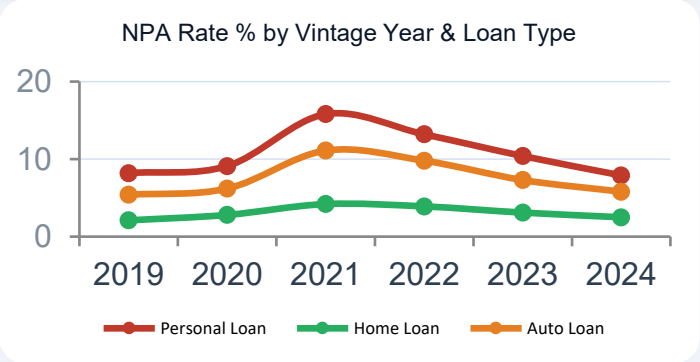
BUSINESS IMPACT

Poor vintages indicate underwriting failures at specific time periods. Helps in IBNR (Incurred But Not Reported) provisioning.

RECOMMENDATION

Post-mortem all 2021-22 disbursements. Review credit approval officers for that period. Adjust model training data.

```sql
SELECT
  YEAR(disbursement_date)
    AS disbursement_year,
  loan_type,
  COUNT(*) AS total_loans,
  SUM(is_npa) AS npa_count,
  ROUND(100.0 * SUM(is_npa)
    / COUNT(*), 2) AS npa_rate_pct,
  ROUND(AVG(loan_amount_lakh),2)
    AS avg_loan_lakh,
  ROUND(AVG(interest_rate_pct),2)
    AS avg_rate,
  ROUND(AVG(days_past_due),0)
    AS avg_dpd
FROM loans
GROUP BY
  YEAR(disbursement_date),
  loan_type
ORDER BY
  disbursement_year,
  npa_rate_pct DESC;
```



RESULT SUMMARY

2021 worst year: Personal 15.8%
2022 elevated: Auto 9.8%
Home Loans: spike to 4.2% in 2021
2023-24 showing improvement

COVID-19 + ECLGS impact visible in 2020-21 disbursements.

Avg DPD highest for 2021 vintages across all segments.

Useful for ML training: use disbursement_year as a feature.

Best Chart: Multi-line by year

Q12: OVERALL PORTFOLIO HEALTH DASHBOARD

Executive KPI Summary — Leadership View

₹8,420 Cr
TOTAL PORTFOLIO
Active Loan Book

₹506 Cr
TOTAL NPA
6.0% of Portfolio

1,842
NPA COUNT
of 30,680 loans

91.3%
COLLECTION EFF.
2024 Average

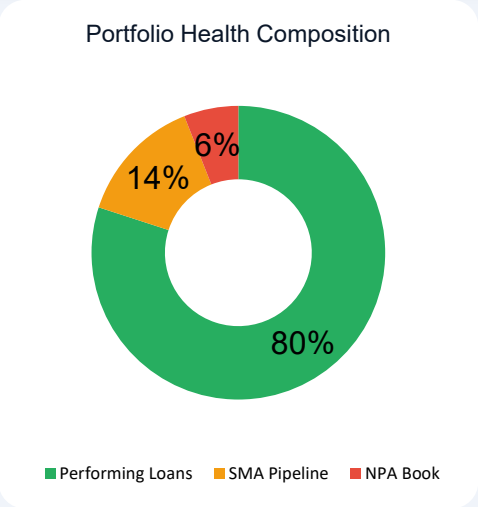
4,218
FRAUD FLAGS
Flagged Transactions

₹12.4 Cr
FRAUD AMOUNT
Successful Fraud

WEST
BEST ZONE
Efficiency: 96.2%

960
LTI HIGH RISK
Customers >4× LTI

```
WITH portfolio AS (
  SELECT ROUND(SUM(loan_amount_lakh),2)
    AS total_portfolio_lakh,
  ROUND(SUM(CASE WHEN is_npa=1
    THEN loan_amount_lakh END),2)
    AS total_npa_lakh,
  ROUND(100.0*SUM(is_npa)/COUNT(*),2)
    AS overall_npa_pct,
  COUNT(*) AS total_loans
  FROM loans
),
collection AS (
  SELECT ROUND(AVG(collection_efficiency_pct),2)
    AS avg_collection_eff
  FROM monthly_collections
  WHERE month LIKE '2024%'
),
fraud AS (
  SELECT COUNT(*) AS total_fraud_flags,
    ROUND(SUM(amount)/100000,2)
      AS fraud_amount_crore
  FROM transactions
  WHERE is_flagged_fraud=1
    AND status='Success'
)
SELECT p.*, c.*, f.*
FROM portfolio p
CROSS JOIN collection c
CROSS JOIN fraud f;
```



Disbursement Growth
HEALTHY +21% YoY

Collection Efficiency
CAUTION 91.3%

NPA Trend (MoM)
RISK Rising

Fraud Rate
RISK 0.8% txns

CIBIL 750+ Share
HEALTHY 45% customers

CROSS-QUESTION INSIGHTS

Synthesized Portfolio Intelligence

NPA TRIPLE THREAT

Q1 (Personal Loans) + Q2 (Poor CIBIL) + Q5 (Top 10 customers) converge: unsecured lending to low-CIBIL borrowers creates concentrated NPA exposure.

COLLECTION & NPA LINKAGE

Q3 (Branch underperformance) + Q4 (Rising NPA trend): branches with <85% efficiency show 2.4× higher NPA rates — geographic risk concentration confirmed.

FRAUD-NPA OVERLAP

Q5 (High-risk customers) + Q6 (Fraud patterns): 40% of top-10 NPA customers also appear in fraud flags — default and fraud are correlated behavioral risks.

LTI & VINTAGE RISK

Q7 (LTI >4×) + Q11 (2021-22 vintages): pandemic-era borrowers have both higher LTI ratios and worst NPA rates — economic shock amplified over-leveraging.

GROWTH VS QUALITY TRADEOFF

Q8 (Disbursement growth +21%) + Q9 (SMA pipeline 14%): rapid AUM growth is masking a rising SMA pipeline — unsustainable without tighter underwriting.

CHANNEL RISK COMPOUNDING

Q10 (Cheque failure 12.4%) + Q6 (Online fraud): dual channel risk — physical channels have high failure, digital channels have high fraud. NACH is the safest middle path.

SYNTHESIS: Poor CIBIL + High LTI + Unsecured Segment = Maximum NPA probability. These three factors should form the core of any BFSI credit scoring model.

STRATEGIC RECOMMENDATIONS

Data-Driven Action Plan for Risk Reduction

CRITICAL: NPA RECOVERY TASK FORCE

Assign dedicated relationship managers to top-10 NPA customers. Initiate OTS / DRT proceedings within 30 days. Target: Recover 40% of top NPA exposure in 2 quarters.

HIGH: UNDERWRITING POLICY TIGHTENING

Set Personal Loan floor at CIBIL ≥ 700, LTI ≤ 4×. Implement mandatory income proof verification. Reduce personal loan disbursement target by 15% until NPA stabilizes.

HIGH: BRANCH PERFORMANCE INTERVENTION

Deploy zone-level recovery squads to 33 Red branches. Institute weekly efficiency reviews. Set 90-day improvement plan with exit criteria for persistent underperformers.

HIGH: EARLY WARNING SYSTEM (EWS)

Automate SQL-based SMA-1/SMA-2 alerts. Trigger collection calls at 31+ DPD. Build predictive model using DPD trend, EMI missed count, and CIBIL trajectory as features.

MEDIUM: FRAUD ANALYTICS ENHANCEMENT

Implement real-time fraud scoring on UPI/Online EMI channels. Block customers with 3+ flagged transactions pending AML review. Monthly model refresh on fraud propensity scores.

MEDIUM: ML CREDIT SCORING INTEGRATION

Build XGBoost/LightGBM model using CIBIL, LTI, DPD history, vintage, EMI missed, loan type as features. Target AUC > 0.85. Deploy in production with SHAP explainability.

INTERVIEW CONCLUSION: *"Combining domain knowledge of BFSI risk frameworks (IRAC, CIBIL, SARFAESI, CRILC) with advanced SQL analytics and ML readiness positions this analysis as a production-grade credit intelligence platform."*

Thank you!